



TEXAS A&M UNIVERSITY
ROBOTICS TEAM & LEADERSHIP EXPERIENCE

Week 3: Transformations in Python

TURTLE Incubator — SO-ARM Software Stack

Python and PIP



Every useful library out there will not be prepackaged with your installation of Python

To install more, we use Python's package manager, **pip**

To install the libraries we need, open your terminal, run `conda activate lerobot`, then type:

```
pip install numpy pandas matplotlib scikit-learn
```

Common Python Libraries



NumPy: Vector/Matrix manipulation

Matplotlib: Building Plots

Pandas: Tables and Spreadsheets

Sci-kit: Statistics and Machine Learning



- NumPy is the go-to library for anything related to vectors
- All the other libraries listed use NumPy under the hood
- Centers around its **ndarray** data structure
- These **ndarrays** have many useful features, such as
 - Transposing/reshaping matrices
 - Slicing the array



What is an Array?



- An array is a data structure for storing and retrieving a list of data
- NumPy's **ndarrays** can have as many dimensions as needed
- **ndarray** is different from Python's **list** in two major ways:
 - Homogeneity: All elements in an array must be of the same data type
 - Fixed size: Once created, the amount of elements in the array cannot change

These restrictions let **ndarrays** be much faster and memory efficient than traditional **lists**

1	5	2	0
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A one-dimensional array of integers

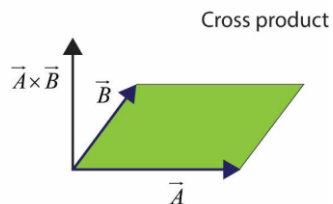
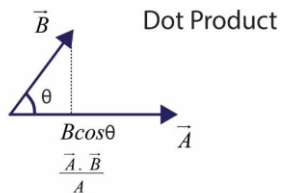
1	5	2	0
8	3	6	1
1	7	2	9

A two-dimensional array of integers

NumPy functionality



You can do many useful linear algebra operations with NumPy arrays like:



Dot and cross product

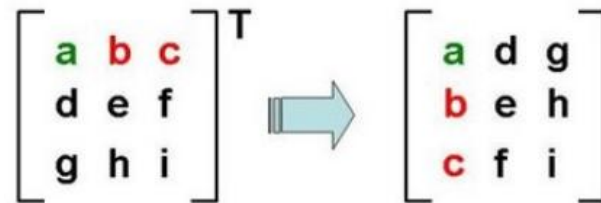
2	3
4	5

`matrix.flatten()`



2	3	4	5
---	---	---	---

Flattening matrices



Transposition/
Reshaping

Code along

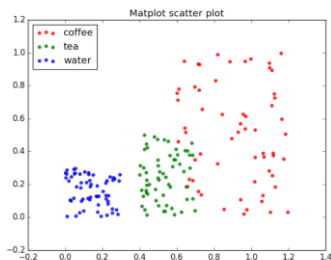


Qr code to starter code
here

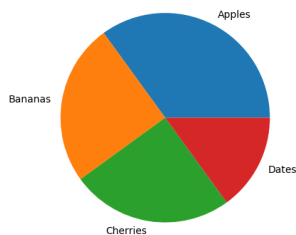


- Matplotlib is a very useful library that allows you to generate many different types of plots.
- Can generate plots from standard lists, NumPy arrays, Pandas tables

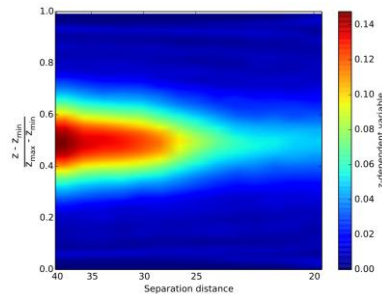
Available plot types



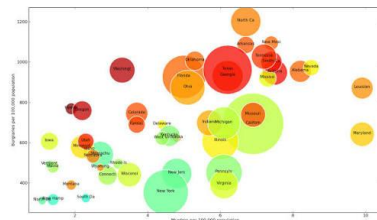
Scatterplots



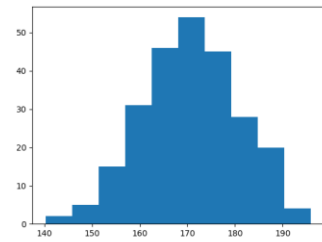
Pie charts



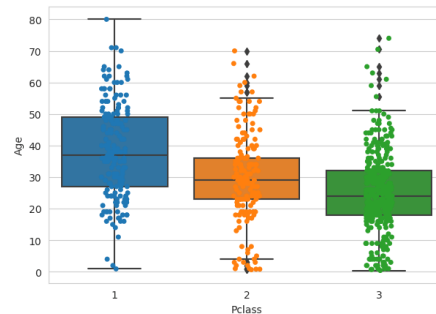
Heatmaps



Bubble plots



Histograms



Box and Whisker plots

And many, many others!



- Pandas is a library for **data manipulation**, particularly when you need to manipulate a spreadsheet-like object (think a super up Microsoft Excel)
- Very useful for cleaning and analyzing data, is the primary tool for data analytics

Key Pandas Features



- Data Importing
 - Can load data from .csv, Excel files, .json, sql, etc.
- Data Cleaning
 - Can quickly find and remove invalid/duplicate data
- Filtering and Slicing
 - Can extract portions of data using conditional logic
 - Like by timestamp, value, etc.
- Merging Datasets
 - Can merge datasets across multiple file formats



- Scikit-learn is the gold-standard machine-learning library for Python
- Note that this is NOT for modern deep learning, more for classical ML like regressions, random forests, clustering

Key Scikit-learn features



- Preprocessing
 - Has many useful functions for normalizing, standardizing, and generally scaling data
- Building models for many forms of ML
 - Classification
 - Identifying which category an object belongs to
 - Ex. Random Forest
 - Regression
 - Predicting a continuous numerical value for an object
 - Ex. Linear regression
 - Clustering
 - Grouping similar objects into larger clusters
 - Ex. K-means clustering

Code along for Matplotlib, Pandas, and Scikit



Idea is to give everyone a .csv file of uncleaned data, have them clean and filter with Pandas, plot it with Matplotlib, then do some linear regression with Scikit.

Qr code to starter code here

Useful properties of matrices



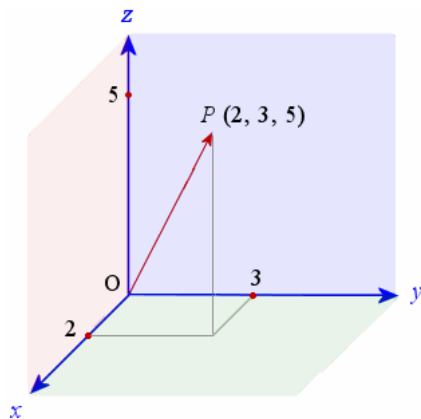
- A matrix is basically just a rectangular 2D array
- Matrices have many useful properties and operations.
- One in particular is **matrix multiplication**
 - We won't go into detail on how to compute a matrix multiplication as you'll usually use **NumPy** to do it
 - (and that's what `ur.linalg` class is for :P)

$$M = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1m} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2m} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3m} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nm} \end{pmatrix}$$

Rotation Matrices



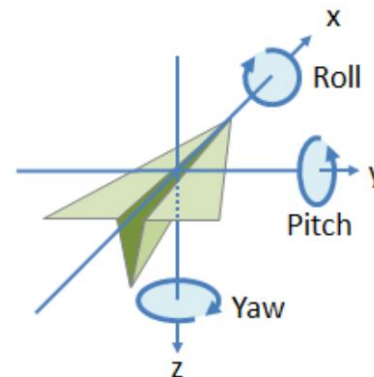
- Rotation Matrices allow you to rotate coordinates by just multiplying the coordinate vector by a matrix
- For example, to rotate the roll, pitch, and yaw of a point/object you can just multiply its coordinate vector by a matrix



$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Picture of Euler Angles

Code Along



Idea is to show how you can compute rotations (either rotation matrices or quaternions), solve systems of equations using linear algebra

QR Code for code along